

Cassini: eBay's Search Engine

JC² Cards Reference Guide — June 2026

1. What Is Cassini?

Cassini is eBay's internal search-ranking algorithm — named after the Saturn probe. When a buyer searches for a card, Cassini decides which of the millions of matching listings appear on page one and which get buried on page twelve. It is the single most important factor in whether your cards get seen at all.

Cassini does not rank by newest or cheapest. It ranks by predicted sale probability — the listings it believes are most likely to result in a completed transaction. Everything else follows from that one idea.

2. The Core Signals Cassini Measures

Cassini scores every listing on a combination of signals. The most important ones for a single-card store like JC² Cards:

Signal	What It Means	Weight
Click-through rate	How often buyers click your listing when it appears in search	Very High
Sell-through rate	What percentage of your listings actually sell	Very High
Price competitiveness	How your price compares to active and recent sold comps	High
Item specifics completeness	Player, Set, Year, Brand, Features fields filled in	High
Title keyword quality	Year + Brand + Player + Set + Parallel in the right order	High
Seller feedback score	Percentage of positive feedback, recency, volume	Medium
Listing age / recency	Newer listings get a brief visibility boost on first appearance	Medium
Photo quality	Clear front + back images; eBay rewards listings with real photos	Medium
Best Offer enabled	Signals buyer-friendliness; Cassini rewards negotiable listings	Low-Med
Promoted listing ad rate	Paid boost on top of organic rank; does not replace organic signals	Additive

3. Price Is the Lever You Control Most

Cassini knows the market. It has access to every completed sale across eBay. If your Cam Ward RC is priced at \$12 and the last 10 sold for \$6, Cassini treats your listing as a likely non-seller and buries it.

The rule of thumb: price within 15% of the recent sold median and Cassini treats your listing as competitive. Price more than 25% above median and your organic rank drops materially. You become invisible.

This is why the daily repricing agent runs every morning. Cards that drift above market get pushed back toward competitiveness before the day's search traffic peaks.

Your Price vs. Sold Median	Cassini Behavior
Within 10% below	Top-tier placement. Maximum impressions.

At median (0%)	Strong placement. Competitive.
10-25% above	Moderate placement. Visible but not featured.
25-50% above	Poor placement. Most buyers never see it.
50%+ above	Effectively invisible in organic search.

4. The June 7 Lesson — How We Learned This the Hard Way

On June 7, 2026, JC² Cards had its best single-day sales volume to date: 9 cards sold. The trigger was a \$1.99 price point on hot rookie chrome parallels (Cam Ward, Travis Hunter, Ashton Jeanty).

The repricing agent had a bug: it re-raised winners back above \$1.99 the next morning. Sales stopped. The fix was two-part: set `max_step_up_pct` to zero to freeze raises, and lock the 8 best-performing impulse cards in a skip list so the agent cannot touch them.

The insight: \$1.99 on a hot rookie chrome is not giving it away. At that price, Cassini ranks it first, buyers click it immediately, and velocity builds. A \$4.99 price on the same card might earn 30% more per unit but sell once a month instead of three times a week. The math favors \$1.99.

5. Item Specifics: The Invisible Work That Matters

Item Specifics are the structured fields on your listing: Brand, Player, Year, Set, Features (RC, Auto, Serial Numbered). Buyers filter by these fields constantly. Cassini uses them to match your listing to searches like “2025 Panini Prizm Cam Ward Rookie” even when those exact words are not in your title.

The store had 354 Panini cards incorrectly labeled as Topps because the posting script had a hardcoded manufacturer field. Those listings were invisible to buyers filtering for Panini products. The `specifics_agent.py` repair job fixes ~23 per pass.

Every listing posted through `post_from_scan.py` now auto-infers Brand, Set, Year, and Features from the title. Never post without a brand token in the title (Prizm, Select, Donruss, Topps Chrome, etc.).

6. Listing Velocity and the Fresh-Listing Boost

New listings receive a 24-48 hour visibility boost when they first go live. Cassini surfaces them to test buyer response — essentially an audition. If buyers click and buy, the listing keeps good placement. If buyers scroll past, Cassini demotes it.

This is why scanning and listing regularly beats dumping 100 cards at once every two weeks. Consistent daily or every-other-day listing keeps the store cycling through fresh-boost windows and signals to Cassini that the store is active and maintaining inventory.

- Aim for 10-30 new listings per session, multiple sessions per week.
- Scan expensive cards first when volume is down — one \$50 auto is worth 25 base cards in Cassini signal value.
- Listings that get early views and watchers get ranked higher permanently.

7. Watcher Offers and Cassini

When a buyer watches a listing, Cassini interprets it as strong buy-intent. Sending an Offer to Watchers (through the Negotiation API) converts that intent into a sale. Cassini then registers the sell-through, which improves the store's overall rank for similar cards.

The optimal timing is not 9am or 5pm. Based on June 2026 data:

- Morning blast (8-10am ET): Catches Friday/weekend lookers starting their day.
- Evening blast (7-9pm ET): Highest conversion window. People are settled, not commuting or working.
- Avoid 5pm ET: Friday afternoon people are leaving work, picking up kids, running errands. Not checking eBay.

8. What Kills Your Cassini Rank

Action	Why It Hurts
Pricing 30%+ above comps	Cassini sees low sale probability and buries the listing
Missing item specifics	Listing misses filtered searches; lower match rate = lower rank
Wrong manufacturer stamped	354 Panini cards labeled Topps — invisible to Panini-filtered searches
Duplicate listings	Cassini penalizes duplicate titles; can suppress both listings
No returns policy edge case	ReturnsNotAccepted is fine; inconsistent return policies confuse the algo
Inactive store periods	No new listings for 2+ weeks signals a dormant store; rank decays
Low sell-through rate	If <5% of your listings sell per month, Cassini demotes the whole store

9. Daily Routine Checklist

The daily pipeline runs these in order to maintain Cassini health:

Step	Agent / Action	Purpose
1	card_price_agent.py	Refresh SportsCardsPro cache
2	repricing_agent.py --apply	Push prices back toward market median
3	best_offer_agent.py --apply	Ensure Best Offer is enabled on all listings
4	watchers_offer_agent.py --apply	Convert watchers to buyers
5	promoted_listings_agent.py --apply	Maintain ad rates on key cards
6	Scan + post new cards	Trigger fresh-listing boosts, add inventory
7	daily_digest_agent.py	Review what sold, what's stale, what to reprice